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The Tyrosine Phosphatase SHP2 Regulates Sertoli Cell Junction Complexes¹

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ABSTRACT

The blood-testis barrier (BTB) is a large junctional complex composed of tight junctions, adherens junctions, and gap junctions between adjacent Sertoli cells in the seminiferous tubules of the testis. Maintenance of the BTB as well as the controlled disruption and reformation of the barrier is essential for spermatogenesis and male fertility. Tyrosine phosphorylation of BTB proteins is known to regulate the integrity of adherens and tight junctions found at the BTB. SHP2 is a nonreceptor protein tyrosine phosphatase (PTP) and a key regulator of growth factor-mediated tyrosine kinase signaling pathways. We found that SHP2 is localized to Sertoli-Sertoli cell junctions in rat testis. The overexpression of a constitutive active SHP2 mutant, SHP2 Q79R, up-regulated the BTB disruptor ERK1/2 via Src kinase in primary rat Sertoli cells in culture. Furthermore, focal adhesion kinase (FAK), which also supports BTB integrity, was found to interact with SHP2 and constitutive activation of SHP2 decreased FAK tyrosine phosphorylation. Expression of the SHP2 Q79R mutant in primary cultured Sertoli cells also resulted in the loss of tight junction and adherens junction integrity that corresponded with the disruption of the actin cytoskeleton and mislocalization of adherens junction and tight junction proteins N-cadherin, β -catenin, and ZO-1 away from the plasma membrane. These results suggest that SHP2 is a key regulator of BTB integrity and Sertoli cell support of spermatogenesis and fertility.

blood-testis barrier, spermatogenesis, testis

INTRODUCTION

The development of germ cells (spermatogenesis) and male fertility are dependent on somatic Sertoli cells that provide nourishment and mechanical support to germ cells. Sertoli cells also create the blood-testis barrier (BTB) that is required for fertility. The BTB is composed of tight junctions, specialized adherens junctions, and gap junctions that physically divide the seminiferous epithelium into the basal and apical compartments while creating a unique environment for germ cell development [1–3]. The BTB is a dynamic structure that must be periodically opened and then re-formed to allow preleptotene spermatocytes to pass from the basal compartment to the adluminal compartment of the seminiferous tubule. Thus, the

timely opening and closing of the BTB is required to maintain spermatogenesis. Maintaining the BTB is also essential because it forms an immunological barrier that sequesters postmeiotic germ cell antigens from the immune system. When the BTB is dysfunctional, germ cell differentiation and development are arrested [4, 5]. The factors and mechanisms regulating the maintenance and periodic remodeling of the BTB are not fully understood. A number of components of the BTB and signals regulating BTB function have been investigated, and kinase-mediated signaling cascades have been found to be important for maintaining the BTB [6]. However, regulation of the BTB by phosphatases has received only limited study [7].

SHP2 is a nonreceptor tyrosine phosphatase containing two amino-terminal SH2 domains, a central phosphotyrosine phosphatase (PTP) domain, and C-terminal tyrosyl phosphorylation sites. In the absence of growth factors, SHP2 is autoinhibited due to interactions between the SH2 and PTP domains. On growth factor or cytokine stimulation, the SH2 domains of SHP2 interact with phosphorylated tyrosine residues on growth factor receptors [8]. Autoinhibition is then relieved, allowing the phosphorylation and activation of SHP2. Once activated, SHP2 stimulates the PI3K/AKT and Ras-MAPK (ERK) pathways, but SHP2 can act as an activator or inhibitor of the JAK/STAT pathway [9–11]. Although best known as a cytoplasmic-plasma membrane localized signaling factor, SHP2 is also localized to the mitochondria, where it regulates cellular redox balance, and to the nucleus, where SHP2 regulates STAT transcription factors and telomerase reverse transcriptase (TERT) activity [12–14].

SHP2 is encoded by the *Ptpn11* gene. In humans, missense mutations in *Ptpn11* that decrease SHP2 activity result in LEOPARD syndrome that is characterized by heart, lung, ocular, growth, and genitalia abnormalities [15]. Disruption of the *Ptpn11* gene in mice results in embryonic lethality [16]. Studies of conditional SHP2 knockout mice have implicated SHP2 in regulating metabolism, obesity, diabetes-insulin sensitivity, liver regeneration, retinal degeneration, cardiovascular protection, and stem cell-induced proliferation of bone marrow-derived mast cells [17–21].

Mutations that disrupt the autoinhibitory conformation of SHP2 result in constitutive activation of the phosphatase and can cause juvenile leukemias [22]. Thus, SHP2 is a proto-oncogene [22, 23]. Constitutive active SHP2 mutations also result in the juvenile development disorder Noonan syndrome, which includes facial dysmorphism, congenital heart defects, short stature, and male infertility [23, 24]. Cryptorchidism was thought to be the main cause of male infertility in Noonan syndrome patients. However, there is evidence that Noonan syndrome patients with normal testicular descent have degenerating spermatocytes and immature Sertoli cells that produce low levels of inhibin. These findings suggest that misregulation of SHP2-mediated signaling causes fertility defects independent of cryptorchidism [25–27].

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TABLE 1. Antibodies employed, applications, and dilutions.

Antibody	Company	Catalog no.	Application	Dilution
SHP2	Santa Cruz Biotechnology	SC-280	Western blotting	1:500
			Immunofluorescence	1:2000
			Immunohistochemistry	1:2000
P-SHP2 (Tyr-542)	Cell Signaling	3751	Western blotting	1:1000
pERK	Cell Signaling	9101s	Western blotting	1:1000
ERK	Upstate	06-182	Western blotting	1:20 000
FAK	Santa Cruz Biotechnology	SC-558	Western blotting	1:1000
p-FAK	Santa Cruz Biotechnology	SC-11765-R	Western blotting	1:2500
β-catenin	Santa Cruz Biotechnology	SC-7199	Western blotting	1:2500
			Immunofluorescence	1:500
α-catenin	Santa Cruz Biotechnology	SC-7894	Western blotting	1:1000
N-cadherin	Santa Cruz Biotechnology	SC-7939	Western blotting	1:1000
			Immunofluorescence	1:500
Paxillin	Santa Cruz Biotechnology	SC-136297	Western blotting	1:1000
p-paxillin	Santa Cruz Biotechnology	SC-101774	Western blotting	1:500
Occludin	Invitrogen	71-1500	Western blotting	1:500
			Immunofluorescence	1:500
ZO-1	Invitrogen	61-7300	Western blotting	1:500
			Immunofluorescence	1:500
JAM-A	Invitrogen	36-1700	Western blotting	1:500
β-actin	Sigma	A5316	Western blotting	1:10 000
			Immunofluorescence	1:1000
Vimentin	Sigma	V6630	Western blotting	1:10 000
			Immunofluorescence	1:1000

SHP2 expression and function have not been characterized in Sertoli cells or other testis cells. However, in other cell types, SHP2 regulates cell adhesion proteins that in Sertoli cells are needed to maintain the BTB and the attachment of elongated spermatids [28]. There is further support for SHP2 as a candidate regulator of the BTB because SHP2 activates ERK and the activation of ERK is associated with disruption of BTB [29]. SHP2 also interacts with and deactivates focal adhesion kinase (FAK) [30, 31], which is required to maintain tight junctions in Sertoli cells [32]. Inappropriate activation of ERK as occurs in Noonan syndrome patients disrupts tight junctions and adherence junctions in other cell types [33–37]. Furthermore, in Sertoli cells, alterations in the phosphorylation status of transmembrane and adaptor proteins can cause junction-associated proteins to be relocalized to the cytoplasm or diffuse through the plasma membrane away from the junction [28, 38, 39]. In this study, we show that SHP2 is a critical regulator of Sertoli cell junctional complexes.

MATERIALS AND METHODS

Animal Care and Use

Male Sprague-Dawley rats were obtained from Charles River Laboratories (Boston, MA). Animals used in these studies were maintained and euthanized according to the principles and procedures described in the National Institutes of Health's *Guide for the Care and Use of Laboratory Animals*. These studies were approved by the University of Pittsburgh Institutional Animal Care and Use Committee.

Reagents, Antibodies, and Adenoviruses

HGF, EGF, PD98059, and wortmannin were obtained from Sigma (St. Louis, MO), PP2 was acquired from VWR (Radnor, PA), and NSC-87877 was purchased from Tocris Bioscience (Bristol, U.K.).

Purchased antisera, their applications and the dilutions used are summarized in Table 1. PP1γ2 antiserum directed against the C terminal 13 amino acids [40] was a gift from S. Vijayaraghavan (Kent State University). AdSHP2 Q79R was obtained as a kind gift from Jeffrey Robbins (University of Cincinnati) [41]. AdEGFP was obtained from the University of Pittsburgh Vector Core Facility.

Isolation of Primary Sertoli Cells

Sertoli cells were isolated from 20-day-old Sprague-Dawley rats and cultured in serum-free media as described previously [42]. Sertoli cells at this age have completed their proliferation period and are no longer dividing in vivo [43–45]. Briefly, decapsulated testes were digested with collagenase (0.5 mg/ml, 33°C, 12 min) in enriched Krebs-Ringer bicarbonate buffer followed by three washes in enriched Krebs-Ringer bicarbonate medium to isolate seminiferous tubules. Tubules were digested with trypsin (0.5 mg/ml, 33°C, 12 min). An equal volume of Dulbecco modified Eagle medium (DMEM) containing 10% fetal calf serum was added to the Sertoli cells, which were then pelleted (100 × g, 5 min) and resuspended in serum-free medium containing 50% DMEM, 50% Hams F-12, 5 mg/ml insulin, 5 mg/ml transferrin, 10 ng/ml epidermal growth factor, 1 mM sodium pyruvate, 200 units/ml penicillin, and 200 mg/ml streptomycin. Sertoli cells were cultured on Matrigel (BD Bioscience, Franklin Lakes, NJ) coated dishes (32°C, 5% CO₂) or Matrigel coated coverslips in dishes. The cells were washed with PBS on Day 2 and cultured further in serum-free media. The cultures were found to be routinely >95% pure as determined by phase microscopy and alkaline phosphatase staining.

Preparation of Whole Cell Extracts, Membrane-Associated Protein Extracts, Western Blots, and Immunoprecipitations

To prepare testis whole cell lysates, testes isolated from 5- to 60-day-old rats were decapsulated and homogenized in enhanced lysis buffer (ELB; 250 mM NaCl, 0.1% NP40, 50 mM Hepes, pH 7.0, 5 mM EDTA, 0.5 mM dithiothreitol) with a protease inhibitor cocktail, rocked for 15 min at 4°C, and then pelleted (12 000 × g 15 min) to remove cell debris. To prepare whole cell lysates of Sertoli cells, the cells were cultured for 3 days and then incubated in the absence and presence of hormones and/or signaling pathway regulators for 10 min. In some cases, cells were pretreated for 4 h with NSC-87877 or 2 h with PP2, PD98059, or wortmannin. To prepare whole cell extracts for direct analysis by Western immunoblot, cells were washed once with PBS and then lysed on the plates by using boiling Laemmli sample buffer to minimize phosphatase activity. Cell lysates were fractionated by SDS-PAGE, transferred to polyvinylidene difluoride membranes, and incubated with primary antibodies, followed by horseradish peroxidase-conjugated second antibody. The antigen-antibody complex was visualized with Millipore Immobilon Western Chemiluminescent HRP substrate (Millipore Corporation, Billerica, MA). Protein concentrations were not performed prior to SDS-PAGE, but equal loading of samples was confirmed with control antisera, including those against actin and nonphosphorylated proteins.

For immunoprecipitations, Sertoli cells were lysed in ELB buffer and a cocktail of protease and phosphatase inhibitors, sonicated for 10 sec, and

subjected to centrifugation ($12000 \times g$ 15 min). The supernatants (5–20 μ g) were added to 500 μ l of ELB buffer and incubated with antibodies followed by incubation with protein G sepharose. Immunobound material was eluted by boiling in 2 $\times$ Laemmli sample buffer for 5 min and fractionated on 10% SDS-PAGE gels. Detection of antigen-bound antibody was carried out by Western analysis as described above.

Immunocytochemistry

Immunostaining of testis tissue was performed on paraffin-embedded sections (5 μ m) from paraformaldehyde fixed (4%, o/n) adult rat testis. Testis sections were deparaffinized in xylene, rehydrated, and then permeabilized for 1 min in cold 100% methanol. The sections were microwaved on high power for 20 min in citrate buffer (10 mM citrate, 30 mM NaCl, pH 5.5) and then left undisturbed at room temperature for 20 min. The sections were washed two times for 5 min in PBS and blocked for 12 h in normal goat serum, 0.5% BSA, and 0.15% glycine at 4°C. Cultured cells were fixed in 4% paraformaldehyde for 5 min, permeabilized for 1 min in ice-cold 100% MeOH, and then dried completely followed by blocking with normal goat serum, 0.5% BSA, and 0.15% glycine. The testis tissue or cultured cells were then incubated 12–24 h with preimmune serum or rabbit polyclonal antiserum directed against SHP2 at a dilution of 1:2000. Anti-rabbit biotinylated secondary antibody (Vectastain Elite ABC Kit, Vector Laboratories, Burlingame, CA) was added, and bound antibodies were detected as described by the kit instructions using DAB staining colorimetric reagent. Slides were washed in H₂O and counterstained with hematoxylin. A charged coupled device video camera system was used to capture images of stained cells or tubule cross sections.

Transepithelial Resistance Assays of Tight Junctions

Sertoli cells isolated from 20-day-old rats were plated onto bicameral units (Millipore Corp, Bedford, MA) coated with Matrigel (BD Bioscience) in 12-well plates at 1.5×10^6 cells/1.1 cm² bicameral unit in serum-free media as previously described [7]. Media was replaced every 24 h. The quality of tight junction formation was assessed by assaying transepithelial resistance (TER) across the Sertoli cell monolayer using a Millicell electrical resistance system (EMB Millipore) as described [7]. Observations were made using duplicate cultures from three different isolations of cultured Sertoli cells.

Immunofluorescence

For confocal studies, rat testes were fixed and sections subjected to antigen retrieval, permeabilized, and blocked as described for the immunocytochemistry studies. The sections were probed with rabbit SHP2 and mouse vimentin antisera followed by incubation with Cy3 and Alexa 488 fluorescently tagged secondary antisera and stained with 4',6-diamidino-2-phenylindole (DAPI). Primary Sertoli cells were cultured in 35-mm² plates containing coverslips in serum-free media as described. Three days after initiating the cultures, the cells were infected with AdSHP2 Q79R or AdGFP. After a further 3 days, the cells were washed twice in ice-cold PBS and fixed with 2% paraformaldehyde in saline for 20 min, washed three times with PBS, blocked with goat serum, and then incubated with a rabbit antiserum directed against SHP2, N-cadherin, ZO-1, or β -catenin or mouse antiserum against β -actin. The cells were washed and then incubated with Cy3 tagged fluorescent secondary goat anti-rabbit antiserum or Cy3 tagged goat anti-mouse antiserum and then stained with DAPI. Images were obtained on a Nikon Provis II (Nikon, Melville, NY) or confocal microscope. All image files were digitally processed for presentation in Adobe Photoshop (Adobe Systems, Inc., San Jose, CA).

Statistical Analysis

Immunoreactive signals from western blot films were scanned with an Epson 1600 Expressions scanner using Epson Scan software. For Western blots, the mean $\pm$ SEM relative signal intensities were determined for at least three independent experiments. Results were analyzed by analysis of variance with Newman-Keuls PLSD or *t*-test at a 5% significance level utilizing GraphPad Prism 4.3 (GraphPad Software, San Diego, CA).

RESULTS

SHP2 Expression in the Testis

Because SHP2 expression has not been characterized previously in the testis, we first studied the developmental expression of SHP2 in postnatal testes. Western blot analysis of

whole cell extracts isolated from rats identified a single immunoreactive protein band at 72 kDa corresponding to SHP2 throughout development. Overall SHP2 expression levels in the testis were relatively constant from 5 to 60 days after birth except for a decrease in signal at 60 days (Fig. 1A). In contrast, the levels of the germ cell-specific PP1 γ 2 protein [40] increased between 16 and 21 days after birth and remained elevated due to the expansion of germ cells that occur during this developmental stage [46].

To determine whether SHP2 expression changes during Sertoli cell development, SHP2 levels were assayed in cultured Sertoli cells isolated from rats 5–20 days after birth. These studies demonstrated that SHP2 was expressed in Sertoli cells and that SHP2 levels remained constant during Sertoli cell development (Fig. 1B). Immunohistochemical analysis of testis sections from an adult rat revealed that SHP2 was expressed in Sertoli cell nuclei, the Sertoli cytoplasm between germ cells extending to the tubule lumen, and that SHP2 was specifically concentrated near the basement membrane in the vicinity of the BTB (Fig. 1, C and D). SHP2 was expressed in spermatogonia, but less SHP2 immunoreactivity was detected in the more mature spermatocytes and spermatids. The region of the Sertoli cell surrounding elongated spermatids also displayed intense SHP2 immunostaining, raising the possibility that SHP2 regulates the attachment of elongated spermatids to Sertoli cells.

Further analysis of SHP2 immunoreactivity using confocal fluorescent microscopy confirmed that SHP2 was expressed in Sertoli cell nuclei as well as Sertoli cytoplasm, including a line of SHP2 immunoreactivity on the apical side of preleptotene spermatocytes and spermatogonia consistent with the position of the BTB (Fig. 1E). SHP2 immunoreactivity in the region of the BTB did not colocalize with vimentin, an intermediate filament protein that is not associated with tight junctions of the BTB. In the Sertoli cell cytoplasm extending to the lumen, SHP2 did colocalize with vimentin, which is known to be expressed in the cytoplasmic extensions of Sertoli cells [47]. Analysis of the expression of SHP2 in Sertoli cells cultured from 20-day-old rats confirmed that SHP2 was expressed in most Sertoli cell nuclei, throughout the cytoplasm, and at the periphery of the cell at cell attachment sites (Fig. 1F).

SHP2 Is Activated by Hepatocyte Growth Factor

Hepatocyte growth factor (HGF), is also known as scatter factor because it disrupts cell-cell adhesion and causes contiguous epithelial cells to disperse [48]. HGF activates SHP2 in other cell types [49] and modulates Sertoli cell tight junctions that are required to maintain the BTB [50], but it is not known whether HGF activates SHP2 in Sertoli cells. To determine whether HGF regulated SHP2 activity, cultured Sertoli cells were stimulated with HGF for 15 min and SHP2 activity was assessed by assaying the extent of phosphorylation at Tyr-542. HGF was found to increase SHP2 and ERK phosphorylation (Fig. 2A). The activation of ERK by HGF was found to be dependent on SHP2 because the SHP2 selective inhibitor NSC-87877 [51] blocked HGF-dependent ERK phosphorylation (Fig. 2B).

SHP2 Regulates MAP Kinase and FAK Signaling in Sertoli Cells

Gain-of-function mutations in SHP2 that cause Noonan syndrome are associated with male infertility and Sertoli cell dysfunction. To determine whether similar overactivation of SHP2 alters the activity of kinases that are essential for Sertoli

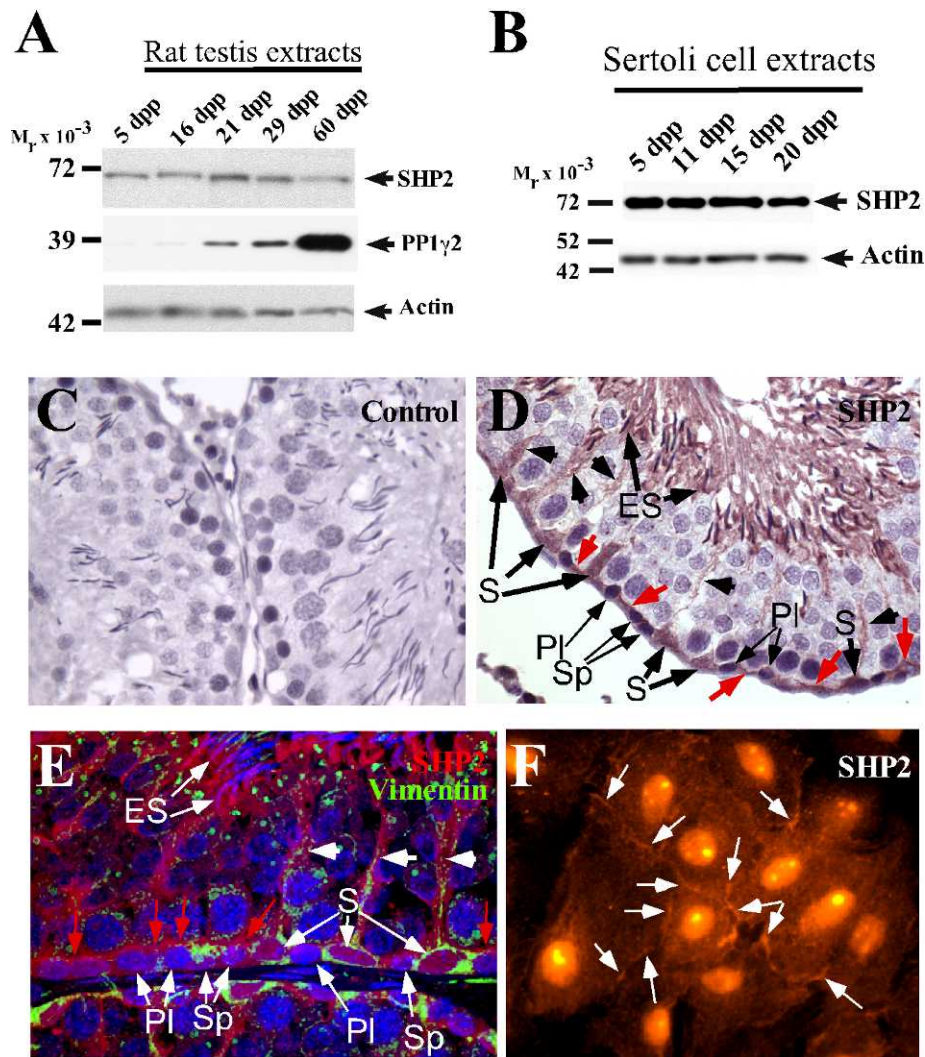


FIG. 1. SHP2 is expressed in Sertoli cells and early stage germ cells in the testis. **A)** Whole testis extracts from rats 5, 16, 21, 29, and 60 days postpartum (dpp) were subjected to Western blot analysis with antiserum against SHP2, PP1 γ 2, or actin. **B)** Whole cell extracts from cultured Sertoli cells isolated from rats 5, 11, 15, and 20 dpp were analyzed by Western blot using antiserum against SHP2 or actin. Adult rat testis sections were probed with secondary antibody only (**C**) or SHP2 antiserum (**D**) and stained with hematoxylin. SHP2 staining (brown) was localized to Sertoli (S) nuclei as well as Sertoli cell cytoplasm between germ cells (black arrowheads) and around elongated spermatids (ES). SHP2 immunostaining was also present along the basement membrane in the region of the blood-testis barrier (red arrows) that did not extend beyond the apical side of spermatogonia (Sp) and preleptotene spermatocytes (PI). **E)** Confocal fluorescence microscopy analysis of SHP2 (red) and vimentin (green) shows that SHP2 is localized to Sertoli nuclei (S) and Sertoli cytoplasm between germ cells (white arrowheads). SHP2 was also detected (red arrows) just to the apical side of preleptotene spermatocytes (PI) and spermatogonia (Sp). SHP2 staining around elongated spermatocytes (ES) is identified by arrows. Nuclei are stained blue with DAPI. **F)** Sertoli cells cultured from 20-day-old Sertoli cells were probed with SHP2 antiserum, and immunolocalization was performed using immunofluorescence microscopy. SHP2 localized to the cell periphery at cell-cell contact points is denoted by arrows. Original magnification $\times 60$ (**C**, **D**, and **F**), $\times 100$ (**E**).

cell-mediated support of spermatogenesis, an adenovirus expressing a constitutive active SHP2 (AdSHP2 Q79R) [41] was introduced into cultured Sertoli cells. Western blot analysis of whole cell extracts determined that SHP2 Q79R increased ERK phosphorylation fourfold (Fig. 3A). Addition of the Src kinase inhibitor PP2 decreased ERK phosphorylation induced by SHP2 Q79R by 68% (Fig. 3B). In contrast, wortmannin, an inhibitor of PI3 kinase, did not affect ERK phosphorylation, whereas the MAP kinase inhibitor PD98059 abolished ERK phosphorylation as expected. These data indicate that much of SHP2 signaling to the MAP kinase cascade is dependent on the activation of Src kinases by SHP2.

SHP2 has not been found to directly interact with Src kinases, but SHP2 has been shown to form a complex with paxillin that in turn binds to and regulates Src activity. When paxillin is dephosphorylated, associated Src is phosphorylated

and activated [52, 53]. We found that phosphorylation of paxillin on its regulatory Tyr 118 residue was decreased by 50% in Sertoli cells expressing SHP2 Q79R (Fig. 3C). These data suggest that SHP2 complexes with and dephosphorylates paxillin, allowing the activation of Src.

FAK kinase regulates the phosphorylation of proteins required to maintain the BTB, and SHP2 has been shown to dephosphorylate FAK in other cell types [30, 31, 54]. Adenoviral-mediated expression of SHP2 Q79R in Sertoli cells decreased FAK phosphorylation on the activating Tyr 397 residue by 80% in comparison to cells infected with a control adenovirus (Fig. 4A). Immunoprecipitation studies determined that SHP2 and FAK interact directly (Fig. 4B). These data suggest that SHP2 directly dephosphorylates FAK in Sertoli cells.

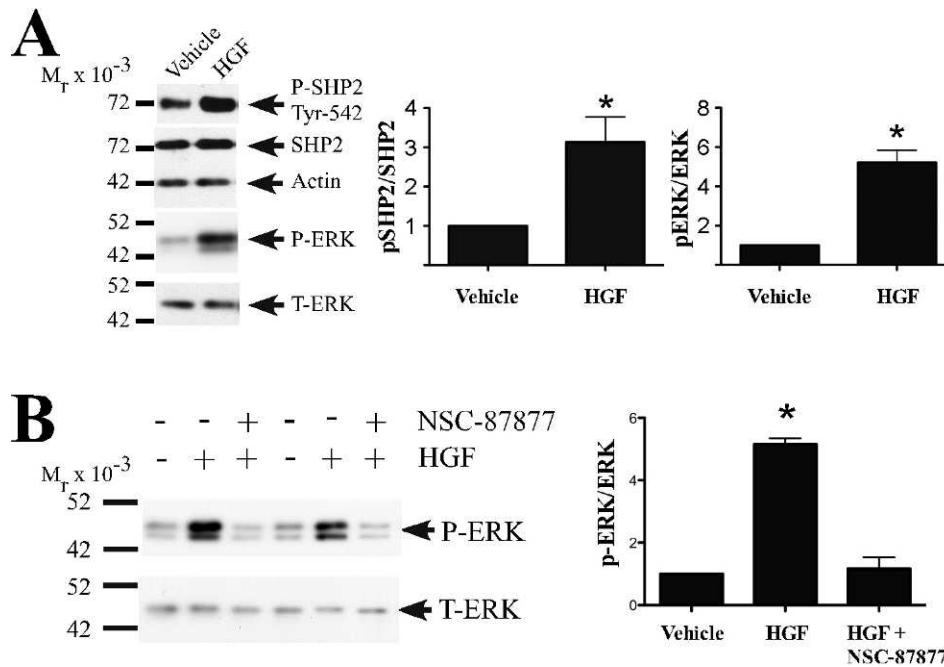


FIG. 2. HGF activates SHP2 and ERK via SHP2. **A**) Cultured Sertoli cells isolated from 20-dpp rats were treated for 15 min with vehicle or HGF (40 ng/ml). Whole cell extracts were subjected to Western blot analysis with antiserum recognizing SHP2 phosphorylated at Tyr-542, total SHP2, actin, P-ERK, and total ERK (T-ERK). **B**) Cultured Sertoli cells were pretreated with vehicle (-) or NSC-87877 for 6 h followed by stimulation with vehicle (-) or HGF (40 ng/ml, +), and whole cell extracts were assayed by Western blot using antiserum against p-ERK and total ERK. *Significant difference from vehicle controls ($P < 0.05$, $n = 3$).

SHP2 Regulates Sertoli Cell Junctional Integrity

The integrity of the BTB is dependent in large part on the formation of tight and adherens junctions between Sertoli cells. To determine whether SHP2 regulates the integrity of tight and adherens junctions formed by Sertoli cells, the cells were cultured as confluent monolayers on bicameral units, and TER was assessed as previously described [7, 55–60]. Infection of cultured Sertoli cells with AdSHP 2Q79R decreased TER by 38%. Stimulation with HGF for 72 h reduced TER levels by 22% (Fig. 4C). These data suggest that continuous stimulation with HGF or extended overactivation of SHP2 inhibits tight junction formation.

To determine whether SHP2 regulates the expression of BTB components, Western blot analysis was performed after adenoviral-mediated expression of SHP2 Q79R or the control GFP protein. These studies indicated that SHP2 Q79R did not alter the expression of the transmembrane proteins N-cadherin, occludin, and JAM-A; the adaptor proteins α -catenin, β -catenin, and ZO-1; or the cytoskeletal protein β -actin (Fig. 5, A–C). However, expression of SHP2 Q79R did alter the localization of the adherens protein N-cadherin and the adaptor proteins ZO-1 and β -catenin. Specifically, after infection of a control adenovirus expressing GFP, the immunofluorescent signals for the proteins were localized predominantly to the periphery of Sertoli cells at adhesion sites associated with the plasma membrane. In contrast, after expression of SHP2 Q79R, the N-cadherin, ZO-1, and β -catenin immunostaining localized at the plasma membrane was decreased (Fig. 6, A–F). Immunofluorescence analysis of β -actin in Sertoli cells also revealed that expression of SHP2Q79R resulted in disruption of the actin cytoskeleton (Fig. 6, G and H). Further coimmunoprecipitation studies revealed that N-cadherin and β -catenin interaction was decreased after overexpression of SHP2 Q79R in cultured Sertoli cells (Fig. 6I). These studies suggest that SHP2 regulates the localization and interactions of

BTB-associated proteins as well as the formation of filamentous actin stress fibers.

DISCUSSION

In this study, we describe the expression, functional significance and potential mechanisms by which SHP2 regulates Sertoli cell adhesion. SHP2 was found to be expressed at a constant level in rat Sertoli cells throughout development, and overall SHP2 levels in the testis do not change from 5 to 29 days after birth. There was a decrease in SHP2 levels between 29 and 60 days after birth that may reflect dilution of the SHP2 signal by the increase in the number of germ cells that occurs with development.

In vivo, we found that SHP2 was expressed in the seminiferous tubules near the basement membrane where the BTB is localized, at sites of Sertoli-cell-elongated spermatid attachment, as well as in the Sertoli cell cytoplasm and nucleus. SHP2 expression also was found to be present in germ cells prior to the preleptotene stage of development, whereas more mature germ cells expressed low levels of SHP2. The strong immunostaining of the Sertoli cell regions associated with the BTB and regions surrounding elongated spermatids supports the hypothesis that SHP2 regulates the maintenance of the BTB and spermatid-Sertoli cell adhesion. SHP2 is a strong candidate for regulating the adherens proteins required to maintain spermatid-Sertoli attachment because SHP2 activates ERK and Src, which in turn phosphorylate proteins required to build and sever the connections between the two cell types [61]. Interestingly, the colorimetric staining of testis sections revealed SHP2 immunostaining near the BTB as a zone extending from the basement membrane to the apical side of spermatogonia and preleptotene spermatocytes, whereas confocal microscopy analysis delineated a more fine line of SHP2 immunoreactivity in the BTB region only on the apical side of spermatogonia and preleptotene spermatocytes. This line of

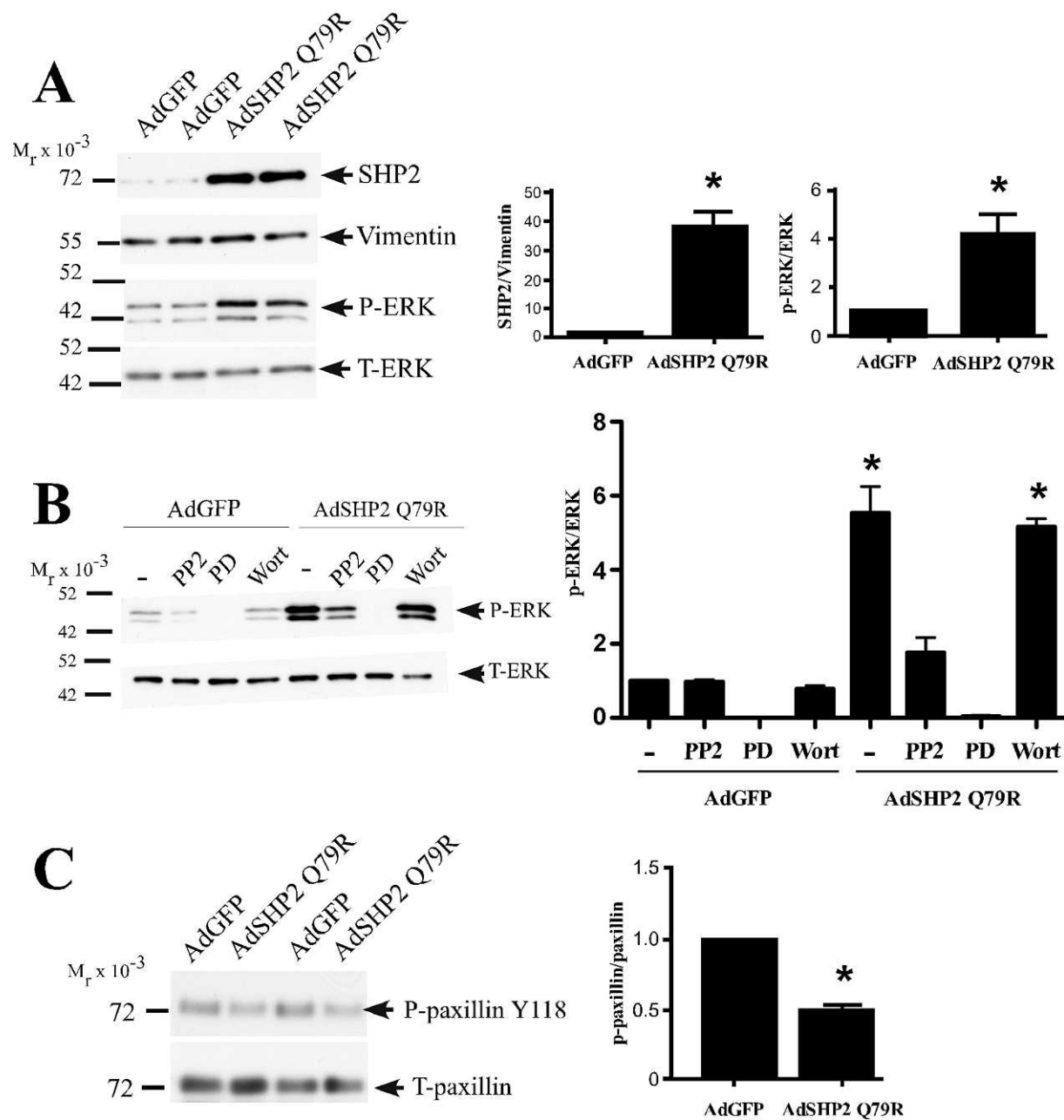


FIG. 3. Constitutively active SHP2 activates ERK via Src kinase and paxillin. **A**) Cultured Sertoli cells from 20-dpp rats were infected in duplicate with AdGFP or AdSHP2 Q79R adenoviruses. Three days after infection, whole cell extracts were assayed by Western blot using antiserum against SHP2, vimentin, P-ERK, or total ERK (T-ERK). Quantitation of SHP2 levels relative to vimentin and P-ERK levels relative to ERK are shown in the graphs to the right. **B**) Cultured Sertoli cells were infected with AdGFP or AdSHP2 Q79R adenoviruses for 3 days and then treated with PP2 (10 nM), PD98059 (50 μ M, PD), or wortmannin (100 nM, Wort) for 2 h, and the levels of P-ERK and T-ERK were determined by Western blot. Quantitation of p-ERK levels relative to total ERK is provided in the graph to the right. **C**) Whole cell extracts from cultured Sertoli cells infected with AdGFP or AdSHP2 Q79R in duplicate for 3 days were probed in Western blot assays with antiserum against paxillin phosphorylated at Y118 and total paxillin. Quantitation of the relative levels of p-paxillin/total paxillin (T-paxillin) is provided in the graph to the right. *Results are statistically different from AdGFP controls ($P < 0.05$, $n = 3$).

SHP2 immunoreactivity is reminiscent of that observed previously for the ZO-1, N-cadherin, and Arp3 proteins associated with the BTB [62]. It is not yet known whether the more diffuse SHP2 immunoreactivity detected by the colorimetric protocol indicates that SHP2 functionality extends from the basement membrane to the BTB or whether the confocal approach provides a higher-resolution result.

HGF stimulation increased SHP2 activity in cultured Sertoli cells. Previously, HGF was shown to activate SHP2 in myoblasts, satellite cells, vascular cells, and pancreatic acinar cells [63–65]. HGF interacting with its receptor c-met results in

the recruitment of Gab1, which then recruits and causes the activation of SHP2 [49]. Interestingly, HGF levels in seminiferous tubules in vivo are low except for a peak during the stages in which the BTB is remodeled to allow the passage of preleptotene spermatocytes through the barrier [66]. Furthermore, HGF was found to modulate the integrity of Sertoli cell tight junctions that are required for the BTB [50]. Although HGF is known to be produced by peritubular myoid cells and Sertoli cells [67], the source of the stage-specific increases in HGF expression is not yet known. Nevertheless, the temporal control of HGF levels in the testis may limit the

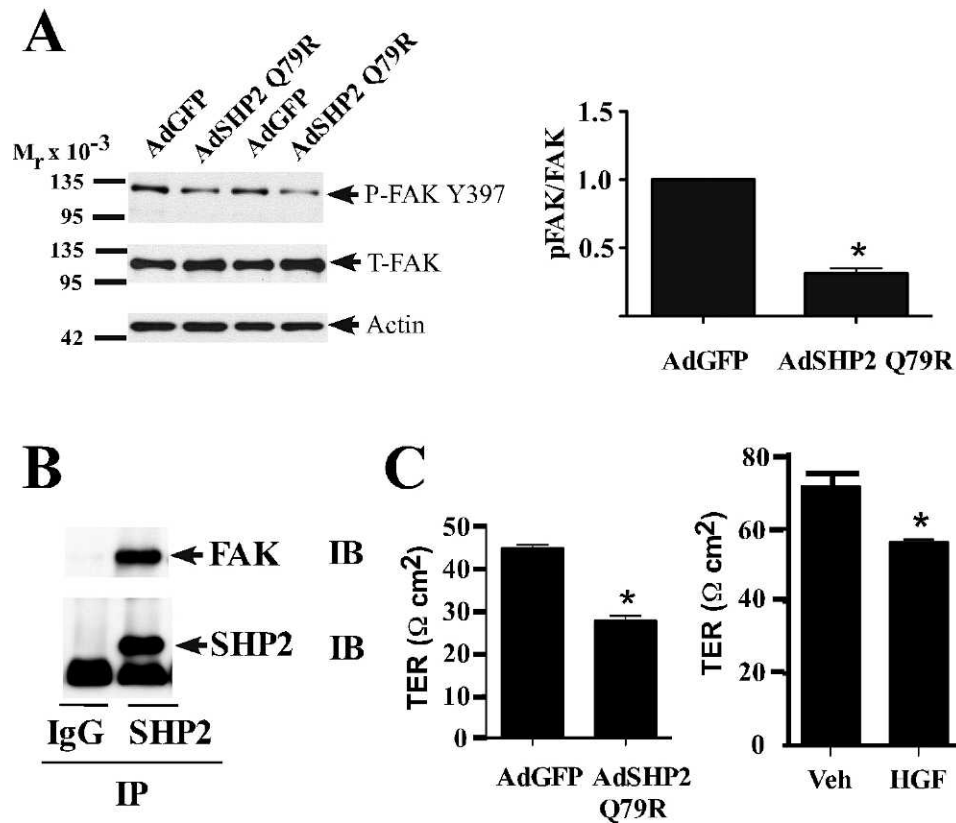


FIG. 4. SHP2 dephosphorylates (inactivates) FAK via a direct interaction. **A**) Cultured Sertoli cells from 20-dpp rats infected in duplicate with control adenovirus (AdGFP) or AdSHP2 Q79R, and 3 days later whole cell extracts were isolated. Western blot analysis of duplicate extracts was performed with antiserum against FAK phosphorylated at Tyr397 (P-FAK Y397), total FAK (T-Fak), or actin. The relative levels of pFAK/total FAK were quantified in the graph to the right ($n = 4$). **B**) Whole cell extracts of Sertoli cells (20 dpp) were immunoprecipitated (IP) with control IgG or antiserum against FAK, and the immunoprecipitates were immunoblotted (IB) with antiserum against FAK (top) or SHP2 (bottom). Results are representative of three experiments. **C**) Sertoli cells infected on Day 2 of culture with AdGFP or AdSHP2 Q79R or Sertoli cells were stimulated with HGF (40 ng/ml) or vehicle (Veh) for the final 3 days of culture before being assayed for TER on Day 6 of culture. *Significant differences from control ($P < 0.05$, $n = 3$).

time span of SHP2 activation that is required for the brief opening and re-formation of the BTB. SHP2 also is stimulated by other BTB regulators, including the cytokine TNF- α [68–70] and beta 1 integrin [71]. TNF- α is produced in a stage-specific fashion by germ cells prior to BTB remodeling [69, 72] and causes the BTB to be transiently disrupted [73]. Beta 1 integrin, when activated, causes the disassembly of adherens junctions that are essential for maintaining the BTB [74]. Together, these findings raise the possibility that SHP2 is a central transducer of signals in Sertoli cells that regulate BTB integrity.

The SHP2 Q79R mutation and other mutations within the SH2 domain result in the constitutive activation of SHP2 and Noonan syndrome in humans. Transgenic mice bearing the SHP2 Q79R mutation show up-regulation of the MAPK pathway and recapitulate many classical symptoms of Noonan syndrome [41, 75]. We determined that, in Sertoli cells, SHP2 Q79R increases the phosphorylation of ERK via Src kinase. We also observed that constitutive activation of SHP2 in Sertoli cells decreased the levels of phosphorylated paxillin in Sertoli cells. Previous studies have shown that SHP2 can act through paxillin to reverse inhibitory tyrosine phosphorylation of the pool of Src that is complexed with paxillin. Specifically, SHP2 dephosphorylation of paxillin results in the dissociation of the negative regulator C-terminal Src kinase (Csk) that normally phosphorylates and inhibits Src in the paxillin-Src complex [52, 53]. We were unable to detect any changes in the phosphorylation of Src in the whole cell extracts of Sertoli cells

expressing constitutive active SHP2 (data not shown). The inability to detect increased p-Src in our whole cell extracts may be explained by the limiting of activated p-Src to the pool of Src associated with paxillin. Thus, the compartmentalized increases in p-Src levels were not detectable above the basal levels of the remaining Src phosphorylation in Sertoli cells. It is also possible that SHP2 may act via other known mechanisms to activate MAP kinase pathway. Specifically, SHP2 could dephosphorylate and inactivate inhibitors of RAS and the MAP kinase pathway, including the RasGAP and sprouty proteins [76].

Expression of constitutively active SHP2 also resulted in reduced tight junction integrity and mislocalization of adaptor proteins required for Sertoli-Sertoli adhesion. One mechanism by which SHP2 may regulate junctional and adhesion proteins is via FAK. Recent studies have established the role of FAK as a regulator of BTB and the blood-brain barrier [32, 77]. RNA interference-mediated knockdown of FAK in Sertoli cells resulted in the disruption of tight junction and adherens junction integrity [32]. In Sertoli cells, FAK complexes with the tight junction protein occludin and the adaptor protein ZO-1 [78]. Down-regulation of FAK signaling results in disruption of interaction between ZO-1 and occludin [77]. Our studies indicate that SHP2 interacts with FAK and decreases phosphorylation at Tyr397 in Sertoli cells, which is associated with decreased FAK activity [79]. FAK phosphorylated at Tyr397 also is localized to specialized junctions between Sertoli and germ cells called ectoplasmic specializations (ES)

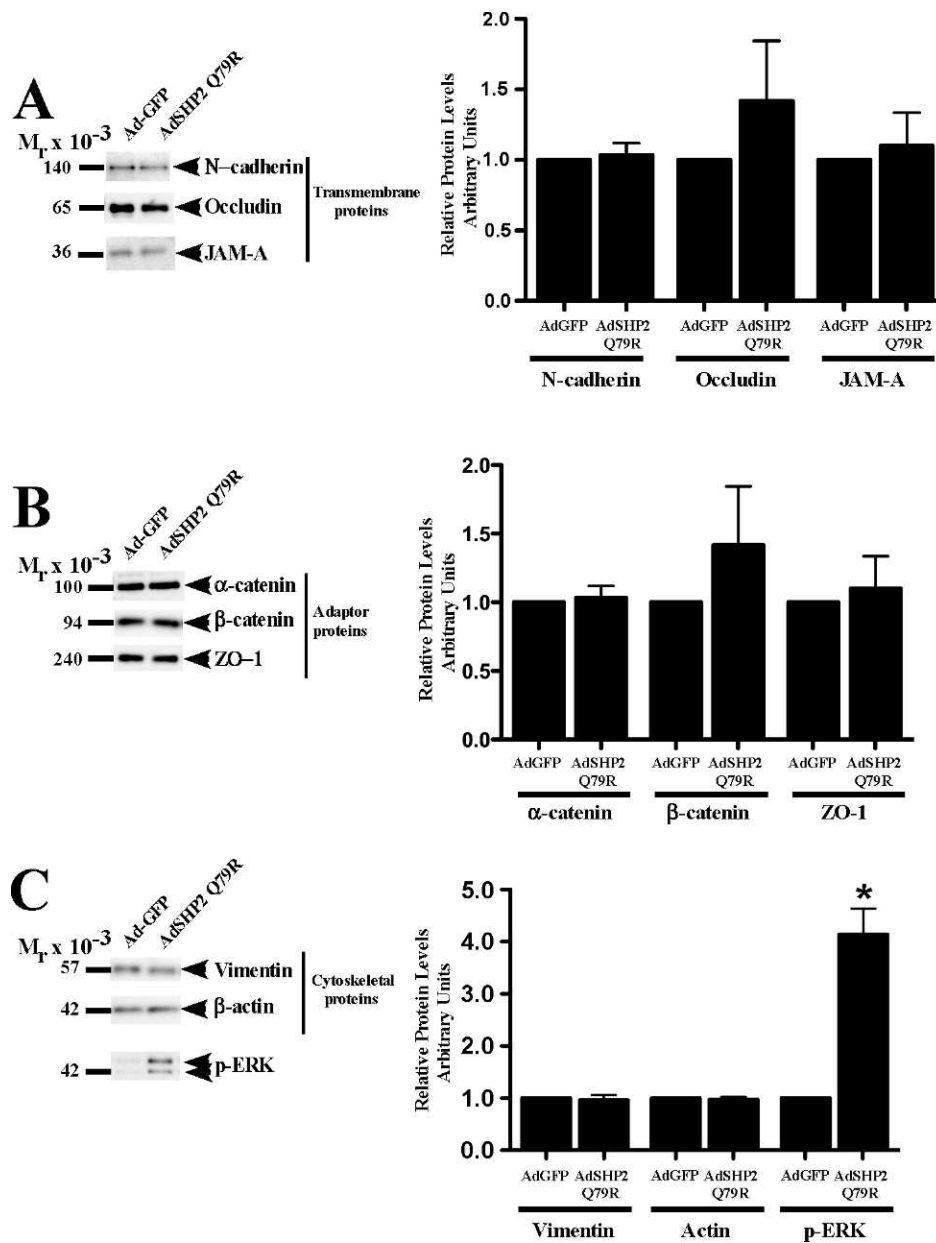


FIG. 5. Constitutively active SHP2 does not alter the expression of transmembrane, adaptor, or cytoskeletal proteins involved in cellular adhesion. Cultured Sertoli cells isolated from 20-day-old rats were infected with AdGFP or AdSHP2 Q79R adenoviruses, and 3 days later Western blots were performed using antiserum against transmembrane proteins (A), adaptor proteins (B), or cytoskeletal proteins (C) as noted. Quantitation of the relative levels of expression for each protein in the presence of AdGFP or AdSHP2 Q79R is shown to the right. *Significant difference from the AdGFP control ($P < 0.05$, $n = 4$).

[80]. Thus, the SHP2 found to localize to Sertoli-elongated spermatid adhesion sites may regulate FAK at the ES.

The decreased TER levels of cultured Sertoli cells observed after expression of SHP2 Q79R indicates that tight junction connections that contribute to the BTB in vivo are decreased after overactivation of SHP2. HGF stimulation of Sertoli cells, which we found increased SHP2 activity and elevated ERK phosphorylation to levels similar to that observed after SHP2 Q79R expression, also significantly decreased TER levels in cultured Sertoli cells. The HGF-mediated decrease in TER levels is consistent with a previous report showing that HGF modulates tight junction dynamics in Sertoli cells by influencing the levels, localization, and assembly of occludin at tight junctions as well as actin organization in the tight junction region [66]. Although these studies indicate that HGF

can disrupt Sertoli cell tight junctions, further work will be required to determine why HGF is not as effective as SHP2 Q79R expression in decreasing tight junction integrity.

Constitutive activation of SHP2 in Sertoli cells caused the mislocalization of adherens and tight junction adaptor proteins. These results are consistent with previously published studies in which misregulation of SHP2 results in disruption of junctional complexes in endothelial cells and fibroblasts [28, 39, 81, 82]. SHP2-mediated disruption of Sertoli cells tight junction and adherens junction integrity could occur by multiple mechanisms initiated after altering the activities of the MAP kinase pathway and Src kinase. Overstimulation of Src and MAP kinase components such as Ras, Raf, and MEK results in disruption of adherens and tight junction integrity [37, 83, 84]. Altering the phosphorylation status of adhesion

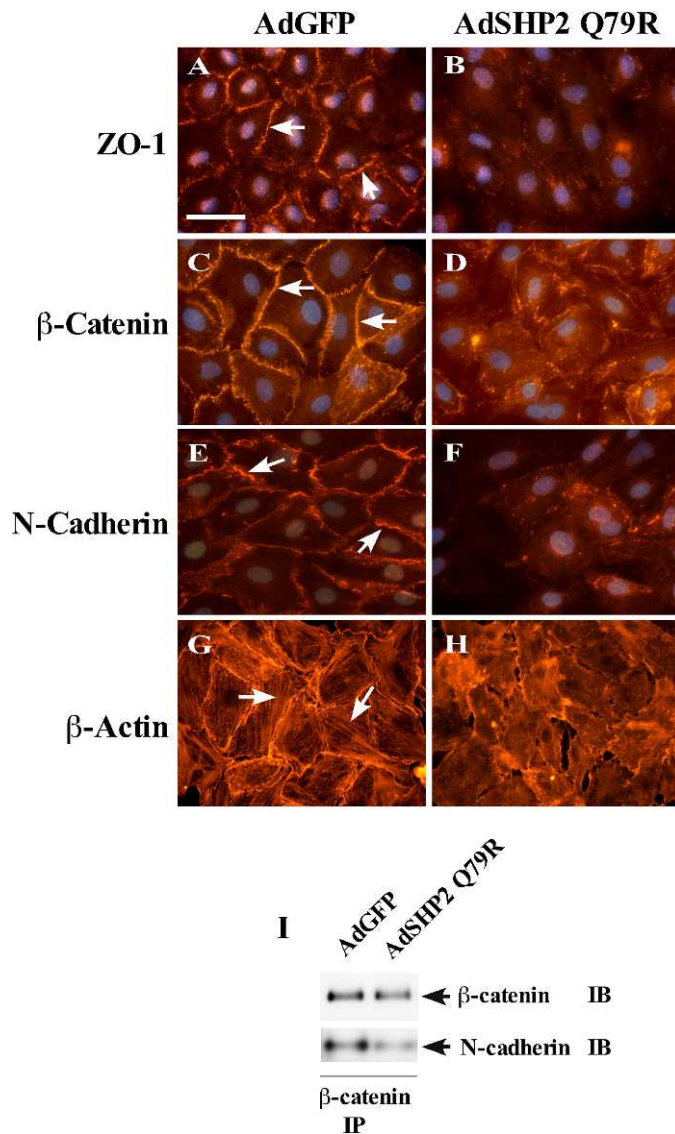


FIG. 6. Constitutive activation of SHP2 results in mislocalization of ZO-1, β -catenin, and N-cadherin away from the plasma membrane and disruption of the actin cytoskeleton. Sertoli cells cultured from 20-day-old rats were infected with AdGFP or AdSHP2 Q79R adenovirus. Three days after infection, the cultures were fixed and subjected to immunofluorescence analysis (red staining) with antiserum against ZO-1 (A and B), β -catenin (C and D), N-cadherin (E and F), or actin (β -actin; G and H). Arrows denote immunostaining at cell-cell contact points (A, C, and E) or actin filaments (G). I) Whole cell extracts from cultured Sertoli cells were immunoprecipitated (IP) with antisera against β -catenin and then immunoblotted with antisera against β -catenin or N-cadherin. Original magnification $\times 60$ (A–H).

proteins is a central mechanism by which the integrity of junction complexes is regulated. Src is known to phosphorylate occludin at Tyr398 and Tyr402 and prevent its interaction with ZO-1, resulting in disrupted tight junctions [85]. Tyrosine phosphorylation of adhesion protein adaptors β -catenin and α -catenin downstream of SHP2 also has been found to disrupt cell adhesion [28, 39, 86]. Alteration of protein phosphorylation at the BTB after expression of SHP2 Q79R in Sertoli cells may explain the decreased interactions of β -catenin with the adherens protein N-cadherin that we observed. This idea is supported by the findings that in other tissues, SHP2 dephosphorylates the adherens junction adaptor protein α -catenin, causing the transmembrane protein cadherin and

adaptor β -catenin to be relocalized to the cytoplasm, resulting in the disruption of the adherens junction [28, 38, 39]. Because SHP2 acts upstream of Src, ERK, FAK, and Rho kinases, it is possible that all of these signaling pathways contribute to SHP2-mediated disruption of Sertoli cell junctions. However, more research is needed to determine the relative contribution of each pathway.

SHP2 Q79R also disrupted the Sertoli cell actin cytoskeleton and decreased polymerized actin filament formation. The disruption of the actin cytoskeleton removes the anchor required for maintaining the BTB and Sertoli-spermatid adhesion. Our findings are consistent with the observations made by other groups reporting that up-regulation of SHP2 activity in epithelial cells and fibroblasts results in fewer actin filaments [39, 81]. SHP2 is a well-established regulator of actin stress fibers via its ability to modulate activity of Rho kinases [81, 87]. Inhibition of SHP2 activity decreases RhoA activity and increases stress fiber formation [28, 81]. Thus, constitutively active SHP2 may disrupt the actin cytoskeleton in Sertoli cells by directly inhibiting Rho kinase. It is also possible that SHP2 may act via ERK kinase to regulate the actin cytoskeleton, as constitutive activation of K-ras and B-Raf via ERK also inhibits Rho activation, resulting in the disruption of actin cytoskeleton [88, 89].

Our studies indicate that SHP2 is a critical regulator of Sertoli cell junction complexes and the integrity of the BTB. Thus, maintaining the correct levels of SHP2 activity in Sertoli cells is essential for spermatogenesis and male fertility. Our observations also provide important information on the pathogenesis of male infertility in Noonan syndrome patients having constitutively active SHP2 mutations. Cryptorchidism was considered to be the main cause of male infertility in Noonan syndrome patients. However, recent studies have shown that Noonan syndrome patients with normal descent of testes have impaired Sertoli cell function [25]. It is likely that the descended testes of Noonan syndrome patients will have disrupted spermatogenesis due to the lack of BTB integrity, although additional effects of SHP2 overactivation may disrupt other Sertoli cell and germ cell functions.

In summary, SHP2 acts in Sertoli cells to regulate spermatogenesis and male fertility by controlling the localization of Sertoli-Sertoli adhesion proteins. SHP2 may also contribute to the regulation of Sertoli-germ cell adhesion. Further studies are required to determine whether SHP2 also acts in developing germ cells to maintain spermatogenesis.

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REFERENCES

1. Cheng CY, Mruk DD. The blood-testis barrier and its implications for male contraception. *Pharmacol Rev* 2012; 64:16–64.
2. Pelletier RM. The blood-testis barrier: the junctional permeability, the proteins and the lipids. *Prog Histochem Cytochem* 2011; 46:49–127.
3. Mital P, Hinton BT, Dufour JM. The blood-testis and blood-epididymis barriers are more than just their tight junctions. *Biol Reprod* 2011; 84: 851–858.
4. Wong CH, Cheng CY. The blood-testis barrier: its biology, regulation, and physiological role in spermatogenesis. *Curr Top Dev Biol* 2005; 71: 263–296.
5. Toyama Y, Maekawa M, Yuasa S. Ectoplasmic specializations in the

- Sertoli cell: new vistas based on genetic defects and testicular toxicology. *Anat Sci Int* 2003; 78:1–16.
6. Wong CH, Cheng CY. Mitogen-activated protein kinases, adherens junction dynamics, and spermatogenesis: a review of recent data. *Dev Biol* 2005; 286:1–15.
 7. Li JC, Mruk D, Cheng CY. The inter-Sertoli tight junction permeability barrier is regulated by the interplay of protein phosphatases and kinases: an in vitro study. *J Androl* 2001; 22:847–856.
 8. Dance M, Montagner A, Salles JP, Yart A, Raynal P. The molecular functions of Shp2 in the Ras/mitogen-activated protein kinase (ERK1/2) pathway. *Cell Signal* 2008; 20:453–459.
 9. Kandadi MR, Stratton MS, Ren J. The role of Src homology 2 containing protein tyrosine phosphatase 2 in vascular smooth muscle cell migration and proliferation. *Acta Pharmacol Sin* 2010; 31:1277–1283.
 10. Grossmann KS, Rosario M, Birchmeier C, Birchmeier W. The tyrosine phosphatase Shp2 in development and cancer. *Adv Cancer Res* 2010; 106: 53–89.
 11. Xu D, Qu CK. Protein tyrosine phosphatases in the JAK/STAT pathway. *Front Biosci* 2008; 13:4925–4932.
 12. Chughtai N, Schimchowitsch S, Lebrun JJ, Ali S. Prolactin induces SHP-2 association with Stat5, nuclear translocation, and binding to the beta-casein gene promoter in mammary cells. *J Biol Chem* 2002; 277: 31107–31114.
 13. Jakob S, Schroeder P, Lukosz M, Buchner N, Spyridopoulos I, Altschmied J, Haendeler J. Nuclear protein tyrosine phosphatase Shp-2 is one important negative regulator of nuclear export of telomerase reverse transcriptase. *J Biol Chem* 2008; 283:33155–33161.
 14. Jakob S, Altschmied J, Haendeler J. “Shping 2” different cellular localizations—a potential new player in aging processes. *Aging (Albany NY)* 2009; 1:664–668.
 15. Edouard T, Montagner A, Dance M, Conte F, Yart A, Parfait B, Tauber M, Salles JP, Raynal P. How do Shp2 mutations that oppositely influence its biochemical activity result in syndromes with overlapping symptoms? *Cell Mol Life Sci* 2007; 64:1585–1590.
 16. Saxton TM, Henkemeyer M, Gasca S, Shen R, Rossi DJ, Shalaby F, Feng GS, Pawson T. Abnormal mesoderm patterning in mouse embryos mutant for the SH2 tyrosine phosphatase Shp-2. *EMBO J* 1997; 16:2352–2364.
 17. Zhang EE, Chapeau E, Hagihara K, Feng GS. Neuronal Shp2 tyrosine phosphatase controls energy balance and metabolism. *Proc Natl Acad Sci U S A* 2004; 101:16064–16069.
 18. Krajewska M, Banares S, Zhang EE, Huang X, Scadeng M, Jhala US, Feng GS, Krajewski S. Development of diabetes in mice with neuronal deletion of Shp2 tyrosine phosphatase. *Am J Pathol* 2008; 172: 1312–1324.
 19. Bard-Chapeau EA, Yuan J, Droin N, Long S, Zhang EE, Nguyen TV, Feng GS. Concerted functions of Gab1 and Shp2 in liver regeneration and hepatoprotection. *Mol Cell Biol* 2006; 26:4664–4674.
 20. Princen F, Bard E, Sheikh F, Zhang SS, Wang J, Zago WM, Wu D, Trelles RD, Bailly-Maitre B, Kahn CR, Chen Y, Reed JC, et al. Deletion of Shp2 tyrosine phosphatase in muscle leads to dilated cardiomyopathy, insulin resistance, and premature death. *Mol Cell Biol* 2009; 29:378–388.
 21. Zhu HH, Ji K, Alderson N, He Z, Li S, Liu W, Zhang DE, Li L, Feng GS. Kit-Shp2-Kit signaling acts to maintain a functional hematopoietic stem and progenitor cell pool. *Blood* 2011; 117:5350–5361.
 22. Tartaglia M, Niemeyer CM, Fragale A, Song X, Buechner J, Jung A, Hahlen K, Hasle H, Licht JD, Gelb BD. Somatic mutations in PTPN11 in juvenile myelomonocytic leukemia, myelodysplastic syndromes and acute myeloid leukemia. *Nat Genet* 2003; 34:148–150.
 23. Chan RJ, Feng GS. PTPN11 is the first identified proto-oncogene that encodes a tyrosine phosphatase. *Blood* 2007; 109:862–867.
 24. Jorge AA, Malaquias AC, Arnold II, Mendonca BB. Noonan syndrome and related disorders: a review of clinical features and mutations in genes of the RAS/MAPK pathway. *Horm Res* 2009; 71:185–193.
 25. Marcus KA, Sweep CG, van der Burgt I, Noordam C. Impaired Sertoli cell function in males diagnosed with Noonan syndrome. *J Pediatr Endocrinol Metab* 2008; 21:1079–1084.
 26. Nistal M, Paniagua R, Pallardo LF. Testicular biopsy and hormonal study in a male with Noonan's syndrome. *Andrologia* 1983; 15:415–425.
 27. Kelnar CJ. Noonan syndrome: the hypothalamo-adrenal and hypothalamo-gonadal axes. *Horm Res* 2009; 72(suppl 2):24–30.
 28. Grinnell KL, Casserly B, Harrington EO. Role of protein tyrosine phosphatase SHP2 in barrier function of pulmonary endothelium. *Am J Physiol Lung Cell Mol Physiol* 2010; 298:L361–L370.
 29. Xia W, Mruk DD, Lee WM, Cheng CY. Differential interactions between transforming growth factor-beta3/TbetaR1, TAB1, and CD2AP disrupt blood-testis barrier and Sertoli-germ cell adhesion. *J Biol Chem* 2006; 281:16799–16813.
 30. Manes S, Mira E, Gomez-Mouton C, Zhao ZJ, Lacalle RA, Martinez AC. Concerted activity of tyrosine phosphatase SHP-2 and focal adhesion kinase in regulation of cell motility. *Mol Cell Biol* 1999; 19:3125–3135.
 31. Miao H, Burnett E, Kinch M, Simon E, Wang B. Activation of EphA2 kinase suppresses integrin function and causes focal-adhesion-kinase dephosphorylation. *Nat Cell Biol* 2000; 2:62–69.
 32. Siu ER, Wong EW, Mruk DD, Porto CS, Cheng CY. Focal adhesion kinase is a blood-testis barrier regulator. *Proc Natl Acad Sci U S A* 2009; 106:9298–9303.
 33. Webb DJ, Donais K, Whitmore LA, Thomas SM, Turner CE, Parsons JT, Horwitz AF. FAK-Src signalling through paxillin, ERK and MLCK regulates adhesion disassembly. *Nat Cell Biol* 2004; 6:154–161.
 34. Wang Y, Zhang J, Yi XJ, Yu FS. Activation of ERK1/2 MAP kinase pathway induces tight junction disruption in human corneal epithelial cells. *Exp Eye Res* 2004; 78:125–136.
 35. Lipschutz JH, Li S, Arisco A, Balkovetz DF. Extracellular signal-regulated kinases 1/2 control claudin-2 expression in Madin-Darby canine kidney strain I and II cells. *J Biol Chem* 2005; 280:3780–3788.
 36. Mullin JM, Leatherman JM, Valenzano MC, Huerta ER, Verrechio J, Smith DM, Snetselaar K, Liu M, Francis MK, Sell C. Ras mutation impairs epithelial barrier function to a wide range of nonelectrolytes. *Mol Biol Cell* 2005; 16:5538–5550.
 37. Li D, Murny RJ. Oncogenic Raf-1 disrupts epithelial tight junctions via downregulation of occludin. *J Cell Biol* 2000; 148:791–800.
 38. Ukropec JA, Hollinger MK, Salva SM, Woolkalis MJ. SHP2 association with VE-cadherin complexes in human endothelial cells is regulated by thrombin. *J Biol Chem* 2000; 275:5983–5986.
 39. Burks J, Agazie YM. Modulation of alpha-catenin Tyr phosphorylation by SHP2 positively effects cell transformation induced by the constitutively active FGFR3. *Oncogene* 2006; 25:7166–7179.
 40. Chakrabarti R, Kline D, Lu J, Orth J, Pilder S, Vijayaraghavan S. Analysis of Ppp1cc-null mice suggests a role for PP1gamma2 in sperm morphogenesis. *Biol Reprod* 2007; 76:992–1001.
 41. Krenz M, Yutzey KE, Robbins J. Noonan syndrome mutation Q79R in Shp2 increases proliferation of valve primordia mesenchymal cells via extracellular signal-regulated kinase 1/2 signaling. *Circ Res* 2005; 97: 813–820.
 42. Cheng J, Watkins SC, Walker WH. Testosterone activates MAP kinase via Src kinase and the EGF receptor in Sertoli cells. *Endocrinology* 2007; 148: 2066–2074.
 43. Orth JM. Proliferation of Sertoli cells in fetal and postnatal rats: a quantitative autoradiographic study. *Anat Rec* 1982; 203:485–492.
 44. Steinberger A, Steinberger E. Replication pattern of Sertoli cells in maturing rat testis in vivo and in organ culture. *Biol Reprod* 1971; 4: 84–87.
 45. Bortolussi M, Zanchetta R, Belvedere P, Colombo L. Sertoli and Leydig cell numbers and gonadotropin receptors in rat testis from birth to puberty. *Cell Tissue Res* 1990; 260:185–191.
 46. Bellvé AR, Millette CF, Bhatnagar YM, O'Brien DA. Dissociation of the mouse testis and characterization of isolated spermatogenic cells. *J Histochem Cytochem* 1977; 25:480–494.
 47. Zhu LJ, Zong SD, Phillips DM, Moo-Young AJ, Bardin CW. Changes in the distribution of intermediate filaments in rat Sertoli cells during the seminiferous epithelium cycle and postnatal development. *Anat Rec* 1997; 248:391–405.
 48. Stoker M, Gherardi E, Perryman M, Gray J. Scatter factor is a fibroblast-derived modulator of epithelial cell mobility. *Nature* 1987; 327:239–242.
 49. Schaeper U, Gehring NH, Fuchs KP, Sachs M, Kempkes B, Birchmeier W. Coupling of Gab1 to c-Met, Grb2, and Shp2 mediates biological responses. *J Cell Biol* 2000; 149:1419–1432.
 50. Catizone A, Ricci G, Galdieri M. Hepatocyte growth factor modulates Sertoli-Sertoli tight junction dynamics. *J Cell Physiol* 2008; 216:253–260.
 51. Chen L, Sung SS, Yip ML, Lawrence HR, Ren Y, Guida WC, Sebt SM, Lawrence NJ, Wu J. Discovery of a novel shp2 protein tyrosine phosphatase inhibitor. *Mol Pharmacol* 2006; 70:562–570.
 52. Ren Y, Meng S, Mei L, Zhao ZJ, Jove R, Wu J. Roles of Gab1 and SHP2 in paxillin tyrosine dephosphorylation and Src activation in response to epidermal growth factor. *J Biol Chem* 2004; 279:8497–8505.
 53. Zhang SQ, Yang W, Kontaridis MI, Bivona TG, Wen G, Araki T, Luo J, Thompson JA, Schraven BL, Phillips MR, Neel BG. Shp2 regulates SRC family kinase activity and Ras/Erk activation by controlling Csk recruitment. *Mol Cell* 2004; 13:341–355.
 54. Marin TM, Clemente CF, Santos AM, Picardi PK, Pascoal VD, Lopes-Cendes I, Saad MJ, Franchini KG. Shp2 negatively regulates growth in cardiomyocytes by controlling focal adhesion kinase/Src and mTOR pathways. *Circ Res* 2008; 103:813–824.
 55. Kaitu'u-Lino TJ, Sluka P, Foo CF, Stanton PG. Claudin-11 expression and

- localisation is regulated by androgens in rat Sertoli cells in vitro. *Reproduction* 2007; 133:1169–1179.
56. Janecki A, Jakubowiak A, Steinberger A. Effects of cyclic AMP and phorbol ester on transepithelial electrical resistance of Sertoli cell monolayers in two-compartment culture. *Mol Cell Endocrinol* 1991; 82: 61–69.
57. Janecki A, Jakubowiak A, Steinberger A. Regulation of transepithelial electrical resistance in two-compartment Sertoli cell cultures: in vitro model of the blood-testis barrier. *Endocrinology* 1991; 129:1489–1496.
58. Chung NP, Mruk D, Mo MY, Lee WM, Cheng CY. A 22-amino acid synthetic peptide corresponding to the second extracellular loop of rat occludin perturbs the blood-testis barrier and disrupts spermatogenesis reversibly in vivo. *Biol Reprod* 2001; 65:1340–1351.
59. Lui WY, Mruk D, Lee WM, Cheng CY. Sertoli cell tight junction dynamics: their regulation during spermatogenesis. *Biol Reprod* 2003; 68: 1087–1097.
60. Siu MK, Lee WM, Cheng CY. The interplay of collagen IV, tumor necrosis factor- α , gelatinase B (matrix metalloproteinase-9), and tissue inhibitor of metalloproteinases-1 in the basal lamina regulates Sertoli cell-tight junction dynamics in the rat testis. *Endocrinology* 2003; 144: 371–387.
61. Shupe J, Cheng J, Puri P, Kostereva N, Walker WH. Regulation of Sertoli-germ cell adhesion and sperm release by FSH and nonclassical testosterone signaling. *Mol Endocrinol* 2011; 25:238–252.
62. Mok KW, Mruk DD, Silvestrini B, Cheng CY. *rpS6* regulates blood-testis barrier dynamics by affecting F-actin organization and protein recruitment. *Endocrinology* 2012; 153:5036–5048.
63. Li J, Reed SA, Johnson SE. Hepatocyte growth factor (HGF) signals through SHP2 to regulate primary mouse myoblast proliferation. *Exp Cell Res* 2009; 315:2284–2292.
64. Shioyama W, Nakaoka Y, Higuchi K, Minami T, Taniyama Y, Nishida K, Kidoya H, Sonobe T, Naito H, Arita Y, Hashimoto T, Kuroda T, et al. Docking protein *Gab1* is an essential component of postnatal angiogenesis after ischemia via HGF/c-met signaling. *Circ Res* 2011; 108:664–675.
65. Hoffmann KM, Tapia JA, Jensen RT. Activation of *Gab1* in pancreatic acinar cells: effects of gastrointestinal growth factors/hormones on stimulation, phosphospecific phosphorylation, translocation and interaction with downstream signaling molecules. *Cell Signal* 2006; 18:942–954.
66. Catizone A, Ricci G, Caruso M, Ferranti F, Canipari R, Galdieri M. Hepatocyte growth factor (HGF) regulates blood-testis barrier (BTB) in adult rats. *Mol Cell Endocrinol* 2012; 348:135–146.
67. Catizone A, Ricci G, Galdieri M. HGF and postnatal testis development. *Mol Cell Endocrinol* 2005; 241:32–40.
68. Hong F, Nguyen VA, Gao B. Tumor necrosis factor α attenuates interferon α signaling in the liver: involvement of SOCS3 and SHP2 and implication in resistance to interferon therapy. *FASEB J* 2001; 15: 1595–1597.
69. Bode JG, Schweigart J, Kehrmann J, Ehrling C, Schaper F, Heinrich PC, Haussinger D. TNF- α induces tyrosine phosphorylation and recruitment of the Src homology protein-tyrosine phosphatase 2 to the gp130 signal-transducing subunit of the IL-6 receptor complex. *J Immunol* 2003; 171:257–266.
70. Mussig K, Staiger H, Fiedler H, Moeschel K, Beck A, Kellerer M, Haring HU. Shp2 is required for protein kinase C-dependent phosphorylation of serine 307 in insulin receptor substrate-1. *J Biol Chem* 2005; 280: 32693–32699.
71. Goel HL, Fornaro M, Moro L, Teider N, Rhim JS, King M, Languino LR. Selective modulation of type I insulin-like growth factor receptor signaling and functions by beta1 integrins. *J Cell Biol* 2004; 166:407–418.
72. De SK, Chen H-L, Pace JL, Hunt JS, Terranova PF, Enders GC. Expression of tumor necrosis factor- α in mouse spermatogenic cells. *Endocrinology* 1993; 133:389–396.
73. Li MW, Xia W, Mruk DD, Wang CQ, Yan HH, Siu MK, Lui WY, Lee WM, Cheng CY. Tumor necrosis factor $\{\alpha\}$ reversibly disrupts the blood-testis barrier and impairs Sertoli-germ cell adhesion in the seminiferous epithelium of adult rat testes. *J Endocrinol* 2006; 190: 313–329.
74. Siu MK, Mruk DD, Lee WM, Cheng CY. Adhering junction dynamics in the testis are regulated by an interplay of beta 1-integrin and focal adhesion complex-associated proteins. *Endocrinology* 2003; 144: 2141–2163.
75. Nakamura T, Gulick J, Pratt R, Robbins J. Noonan syndrome is associated with enhanced pERK activity, the repression of which can prevent craniofacial malformations. *Proc Natl Acad Sci U S A* 2009; 106: 15436–15441.
76. Neel BG, Gu H, Pao L. The ‘Shp’ing news: SH2 domain-containing tyrosine phosphatases in cell signaling. *Trends Biochem Sci* 2003; 28: 284–293.
77. Su L, Mruk DD, Lui WY, Lee WM, Cheng CY. P-glycoprotein regulates blood-testis barrier dynamics via its effects on the occludin/zonula occludens 1 (ZO-1) protein complex mediated by focal adhesion kinase (FAK). *Proc Natl Acad Sci U S A* 2011; 108:19623–19628.
78. Siu ER, Wong EW, Mruk DD, Sze KL, Porto CS, Cheng CY. An occludin-focal adhesion kinase protein complex at the blood-testis barrier: a study using the cadmium model. *Endocrinology* 2009; 150:3336–3344.
79. Tsutsumi R, Takahashi A, Azuma T, Higashi H, Hatakeyama M. Focal adhesion kinase is a substrate and downstream effector of SHP-2 complexed with *Helicobacter pylori* CagA. *Mol Cell Biol* 2006; 26: 261–276.
80. Wong CH, Xia W, Lee NP, Mruk DD, Lee WM, Cheng CY. Regulation of ectoplasmic specialization dynamics in the seminiferous epithelium by focal adhesion-associated proteins in testosterone-suppressed rat testes. *Endocrinology* 2005; 146:1192–1204.
81. Kodama A, Matozaki T, Fukuhara A, Kikyo M, Ichihashi M, Takai Y. Involvement of an SHP-2-Rho small G protein pathway in hepatocyte growth factor/scatter factor-induced cell scattering. *Mol Biol Cell* 2000; 11:2565–2575.
82. Hatanaka K, Lanahan AA, Murakami M, Simons M. Fibroblast growth factor signaling potentiates VE-cadherin stability at adherens junctions by regulating SHP2. *PLoS One* 2012; 7:e37600.
83. Marschitz I, Lechner J, Mosser I, Dander M, Montesano R, Schramek H. Differential expression of cell-cell adhesion proteins and cyclin D in MEK1-transdifferentiated MDCK cells. *Am J Physiol Cell Physiol* 2000; 279:C1472–C1482.
84. Reuveni H, Geiger T, Geiger B, Levitzki A. Reversal of the Ras-induced transformed phenotype by HR12, a novel ras farnesylation inhibitor, is mediated by the Mek/Erk pathway. *J Cell Biol* 2000; 151:1179–1192.
85. Elias BC, Suzuki T, Seth A, Giorgianni F, Kale G, Shen L, Turner JR, Naren A, Desiderio DM, Rao R. Phosphorylation of Tyr-398 and Tyr-402 in occludin prevents its interaction with ZO-1 and destabilizes its assembly at the tight junctions. *J Biol Chem* 2009; 284:1559–1569.
86. Roura S, Miravet S, Piedra J, Garcia de Herreros A, Dunach M. Regulation of E-cadherin/catenin association by tyrosine phosphorylation. *J Biol Chem* 1999; 274:36734–36740.
87. Schoenwaelder SM, Petch LA, Williamson D, Shen R, Feng GS, Burrige K. The protein tyrosine phosphatase Shp-2 regulates RhoA activity. *Curr Biol* 2000; 10:1523–1526.
88. Klein RM, Spofford LS, Abel EV, Ortiz A, Aplin AE. B-Raf regulation of Rnd3 participates in actin cytoskeletal and focal adhesion organization. *Mol Biol Cell* 2008; 19:498–508.
89. Sahai E, Olson MF, Marshall CJ. Cross-talk between Ras and Rho signalling pathways in transformation favours proliferation and increased motility. *EMBO J* 2001; 20:755–766.